



User-defined functions

```
Python — vi function.py — 84×21
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#User-defined function
#####

def maxVal(x, y):
    if x > y:
        return x
    else:
        return y

int1 = int(input('Enter an integer: '))
int2 = int(input('Enter an integer: '))

max = maxVal(int1, int2)           #positional binding of arguments to formal parameters
print(max, 'is greater')
~
~
~
"function.py" 17L, 418C
```

```
Python — -bash — 51×5
[Deepaks-MacBook-Pro:Python deepak$ ./function.py
Enter an integer: 7
Enter an integer: 2
7 is greater
Deepaks-MacBook-Pro:Python deepak$ ]
```

Keyword Arguments



```
Python — vi keyword_arg.py — 87x20
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#User-defined function and Keyword arguments
#####

def printName(firstName, lastName, reverse):
    if reverse:
        print(lastName + ', ' + firstName)
    else:
        print(firstName, lastName)

printName('Deepak', 'Sharma', False)    #positional binding of arg to formal parameters
printName(lastName='Sharma', firstName='Deepak', reverse=False)    #keyword arguments

#printName(firstName = 'Deepak', lastName = 'Sharma', False)
~
~
~
```

"keyword_arg.py" 16L, 571C

```
Python — -bash — 54x5
Deepaks-MacBook-Pro:Python deepak$ ./keyword_arg.py
Deepak Sharma
Deepak Sharma
Deepaks-MacBook-Pro:Python deepak$
```



Keyword Arguments (Default Parameters)

```
Python — vi default_keyword.py — 87x20
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#User-defined function, Keyword arguments and Default parameters
#####

def printName(firstName, lastName, reverse = False):           #default parameter value
    if reverse:
        print(lastName + ', ' + firstName)
    else:
        print(firstName, lastName)

printName('Deepak', 'Sharma')                                #no need of 3rd argument
printName('Deepak', 'Sharma', reverse = True)                #keyword arg can follow positional arg

#printName(firstName = 'Deepak', lastName = 'Sharma', False)
~
~
~
"default_keyword.py" 16L, 657C
```

```
Python — -bash — 57x5
Deepaks-MacBook-Pro:Python deepak$ ./default_keyword.py
Deepak Sharma
Sharma, Deepak
Deepaks-MacBook-Pro:Python deepak$
```

Scoping



```
Python — vi scoping.py — 70×22
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#User-defined function and scoping
#####

def f(x):                #name x used as a formal parameter
    y = 1
    x = x + y
    print('x =', x)
    return x

x = 2
y = 7

z = f(x)                #value of x used as actual parameter
print('z =', z)
print('x =', x)
print('y =', y)
~
~
"scoping.py" 19L, 408C
```

```
Python — -bash — 48×6
Deepaks-MacBook-Pro:Python deepak$ ./scoping.py
x = 3
z = 3
x = 2
y = 7
Deepaks-MacBook-Pro:Python deepak$
```

Modules



```
Python — vi module.py — 65×19
#!/Library/Frameworks/Python.framework/Version
#####
#User-defined function and modules
#####

import circle

pi = 3

print(pi)
print(circle.pi)
print(circle.area(3))
print(circle.circumference(3))
print(circle.sphereSurface(4))
print(circle.sphereVolume(4))
~
~
"module.py" 16L, 335C
```

```
Python — vi circle.py — 40×15
pi = 3.14159

def area(radius):
    return pi*(radius**2)

def circumference(radius):
    return 2*pi*radius

def sphereSurface(radius):
    return 4.0*area(radius)

def sphereVolume(radius):
    return (4.0/3.0)*pi*(radius**3)
~
"circle.py" 13L, 216C
```

```
Python — -bash — 47×8
Deepaks-MacBook-Pro:Python deepak$ ./module.py
3
3.14159
28.27431
18.849539999999998
201.06176
268.08234666666664
Deepaks-MacBook-Pro:Python deepak$
```

print (f)



- Utilizes *f-strings* (formatted string literals) for embedding expressions inside string literals, evaluated at runtime

```
In [1]: name = "Deepak"  
age = 48  
print(f"My name is {name} and I am {age} years old.")
```

My name is Deepak and I am 48 years old.

```
In [2]: height = 174.75  
print(f"My height is {height:.1f} cm")
```

My height is 174.8 cm

Handling exceptions

```
In [1]: def divide_numbers(x, y):  
        try:  
            result = x / y  
            print("The result is:", result)  
        except ZeroDivisionError:  
            print("Error: Cannot divide by zero!")  
        except Exception as e:  
            print(f"An error occurred: {e}")  
        finally:  
            print("This always executes.")
```

```
In [2]: divide_numbers(10, 0)
```

```
Error: Cannot divide by zero!  
This always executes.
```

```
In [3]: divide_numbers(10, 'a')
```

```
An error occurred: unsupported operand type(s) for /: 'int' and 'str'  
This always executes.
```

```
In [4]: divide_numbers(10, 2)
```

```
The result is: 5.0  
This always executes.
```

File writing



```
Python — vi file_writing.py — 67×15
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Writing a file
#####

kidsHandle = open('kids', 'w')
for i in range(2):
    name = input('Enter name: ')
    kidsHandle.write(name + '\n')
kidsHandle.close()
~
~
~
"file_writing.py"
```

```
Python — -bash — 57×6
[Deepaks-MacBook-Pro:Python deepak$ ./file_writing.py
Enter name: Abdul
Enter name: Sumesh
Deepaks-MacBook-Pro:Python deepak$
```

```
Python — vi kids...
Abdul
Sumesh
~
~
"kids" 2L, 13C
```

File reading



```
Python — vi file_reading.py — 66×22
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3
```

```
#####
#Reading a file
#####
```

```
kidsHandle = open('kids', 'r')
print('*****')
for line in kidsHandle:
    print(line)
print('*****')
kidsHandle.close()
```

```
kidsHandle = open('kids', 'r')
print('#####')
names = kidsHandle.read()
print(names)
print('#####')
~
~
"file_reading.py" 19L, 436C
```

```
Python — -bash — 54×13
[Deepaks-MacBook-Pro:Python deepak$ ./file_reading.py]
```

```
*****
Abdul
```

```
Sumesh
```

```
*****
#####
Abdul
Sumesh
```

```
#####
```

```
Deepaks-MacBook-Pro:Python deepak$
```

File reading (take 2)



```
Python — vi file_reading2.py — 65x14
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Reading a file (take2)
#####

kidsHandle = open('kids', 'r')
for line in kidsHandle:
    print(line[:-1])
kidsHandle.close()
~
~
~
"file_reading2.py" 10L,
```

```
Python — -bash — 56x5
Deepaks-MacBook-Pro:Python deepak$ ./file_reading2.py
Abdul
Sumesh
Deepaks-MacBook-Pro:Python deepak$
```

File appending



```
Python — vi file_appending.py — 65x17
#!/Library/Frameworks/Python.framework/Versions/3.7/bin/python3

#####
#Writing/Appending a file
#####

kidsHandle = open('kids', 'a')

kidsHandle.write('Shalini\n')

for i in range(2):
    name = input('Enter name: ')
    kidsHandle.write(name + '\n')
kidsHandle.close()

~
~
"file_appending.py" 14L,
```

```
Python — -bash — 55x5
[Deepaks-MacBook-Pro:Python deepak$ ./file_appending.py ]
Enter name: Snigdha
Enter name: Anil
Deepaks-MacBook-Pro:Python deepak$
```

```
Python — vi kids...
Abdul
Sumesh
Shalini
Snigdha
Anil
~
"kids" 5L, 34C
```

Practice Programs



PP3: Make a program to print the following design (take the input of lines from the user):

```
*  
**  
***  
****  
.....
```

PP4: Make a program to find the sum of even and odd numbers till the number specified by the user.

Practice Programs



PP5: *Make a program to reverse a number using a while loop.*